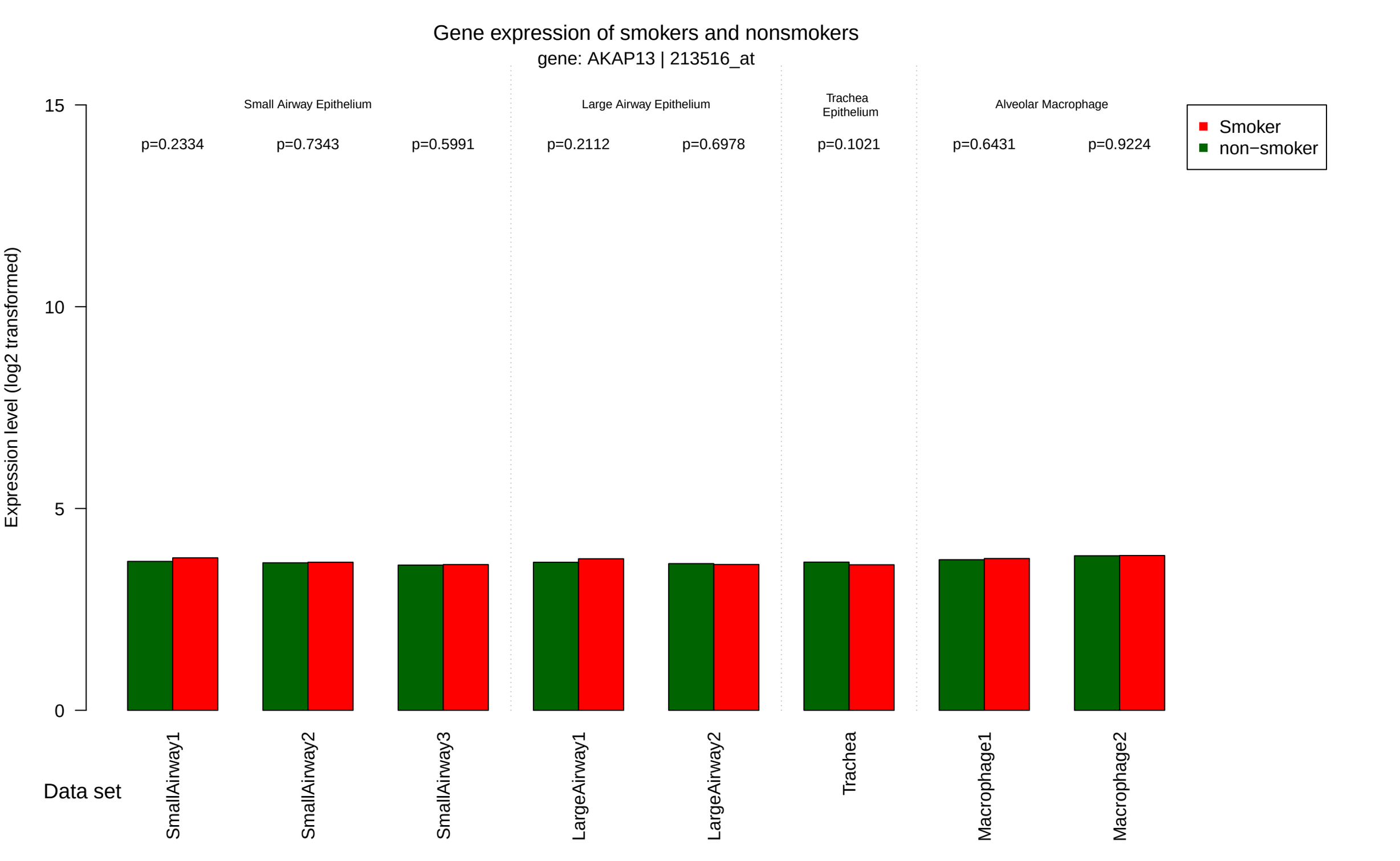
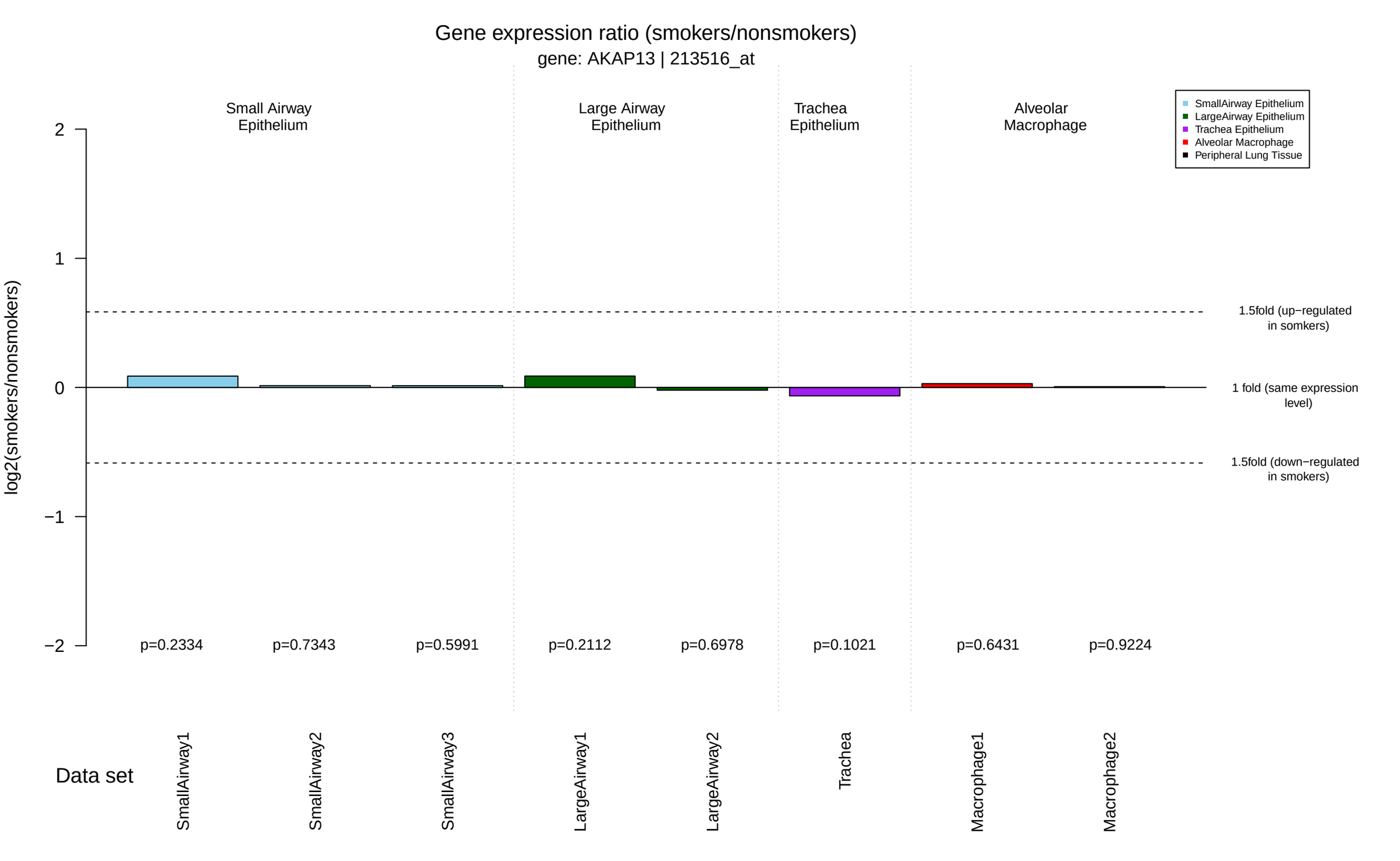
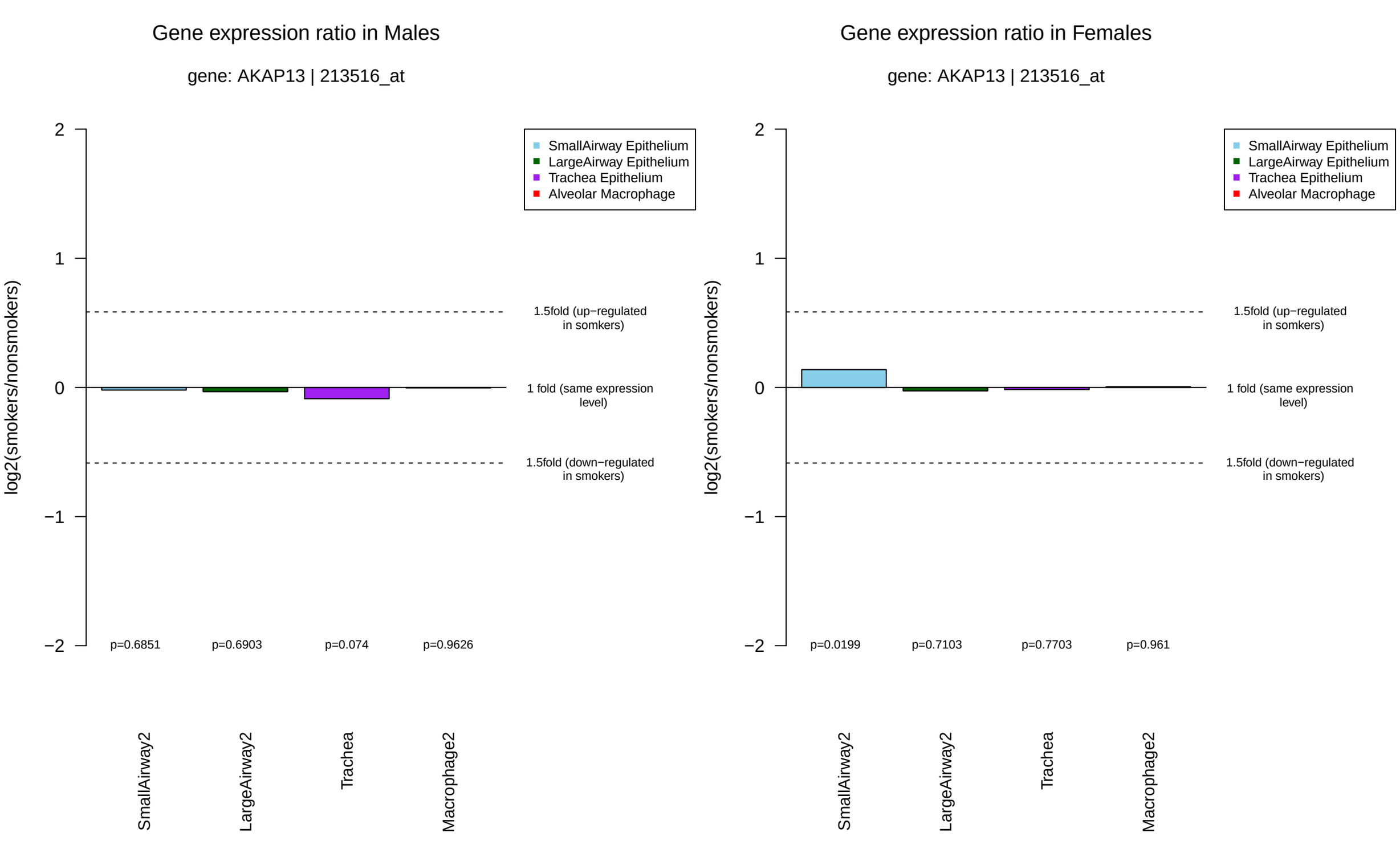
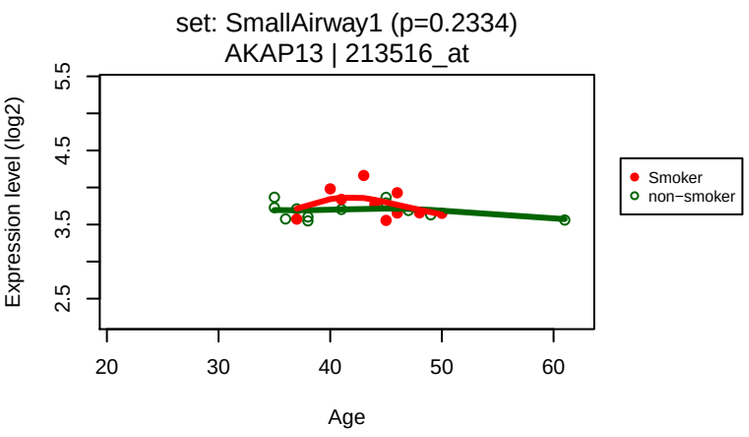
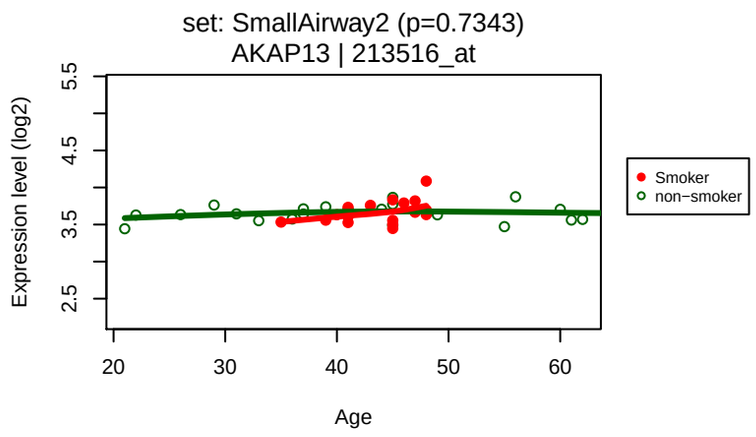
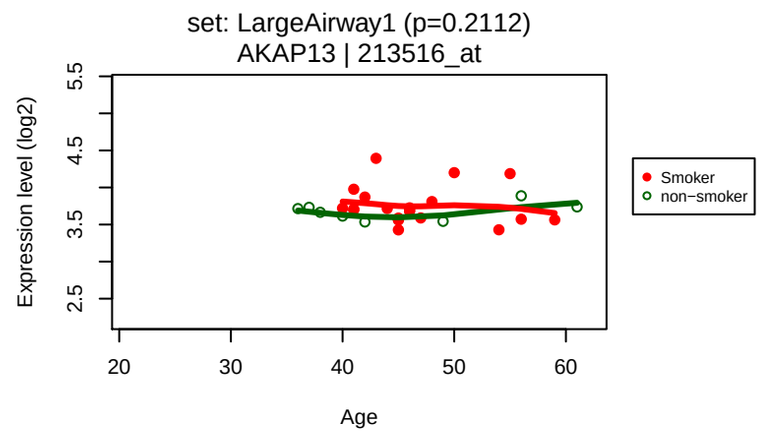

Gene expression of smokers and nonsmokers
gene: AKAP13 | 213516_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2334
p=0.7343
p=0.5991
p=0.2112
p=0.6978
p=0.1021
p=0.6431
p=0.9224
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

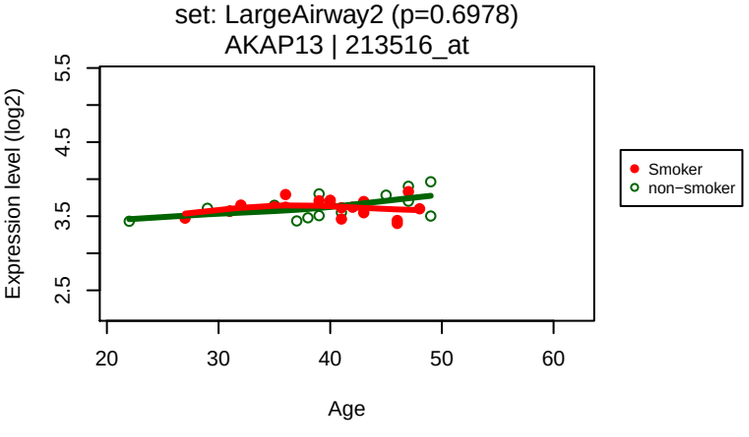

Gene expression ratio (smokers/nonsmokers)
gene: AKAP13 | 213516_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2334
p=0.7343
p=0.5991
p=0.2112
p=0.6978
p=0.1021
p=0.6431
p=0.9224
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

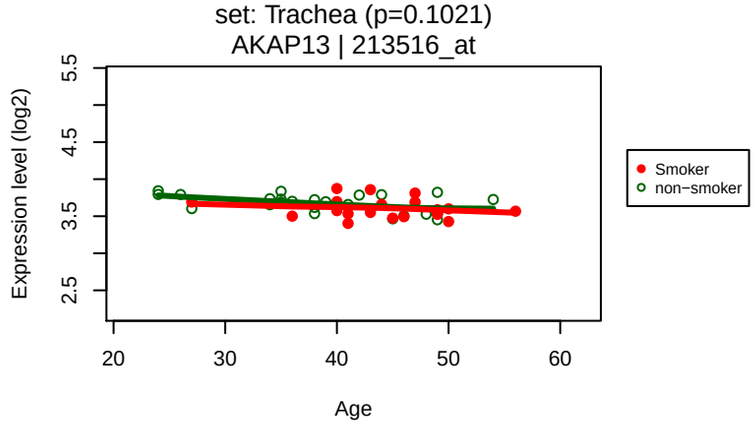

Gene expression ratio in Males
gene: AKAP13 | 213516_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6851
p=0.6903
p=0.074
p=0.9626
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP13 | 213516_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0199
p=0.7103
p=0.7703
p=0.961
SmallAirway2
LargeAirway2
Trachea
Macrophage2

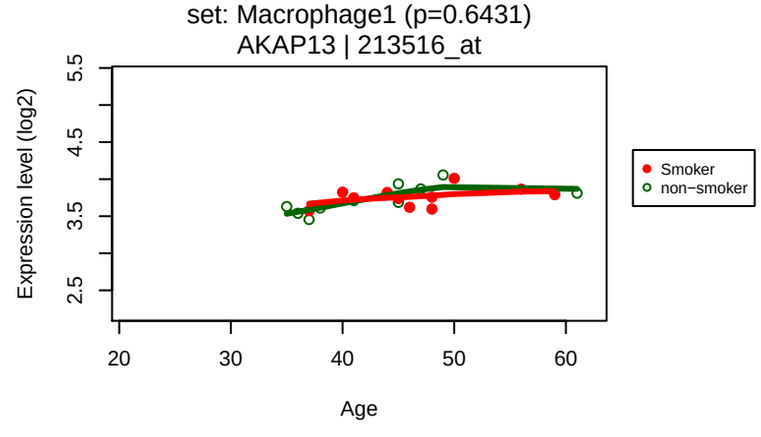

set: SmallAirway1 (p=0.2334)
AKAP13 | 213516_at
Expression level (log2)
Age
Smoker
non−smoker

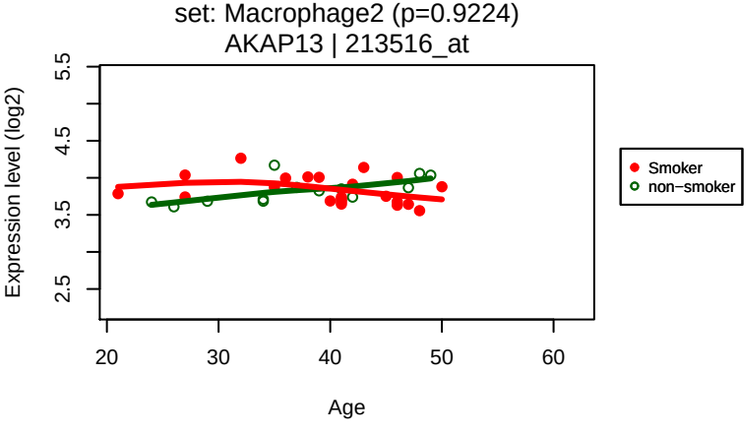

set: SmallAirway2 (p=0.7343)
AKAP13 | 213516_at
Expression level (log2)
Age
Smoker
non−smoker

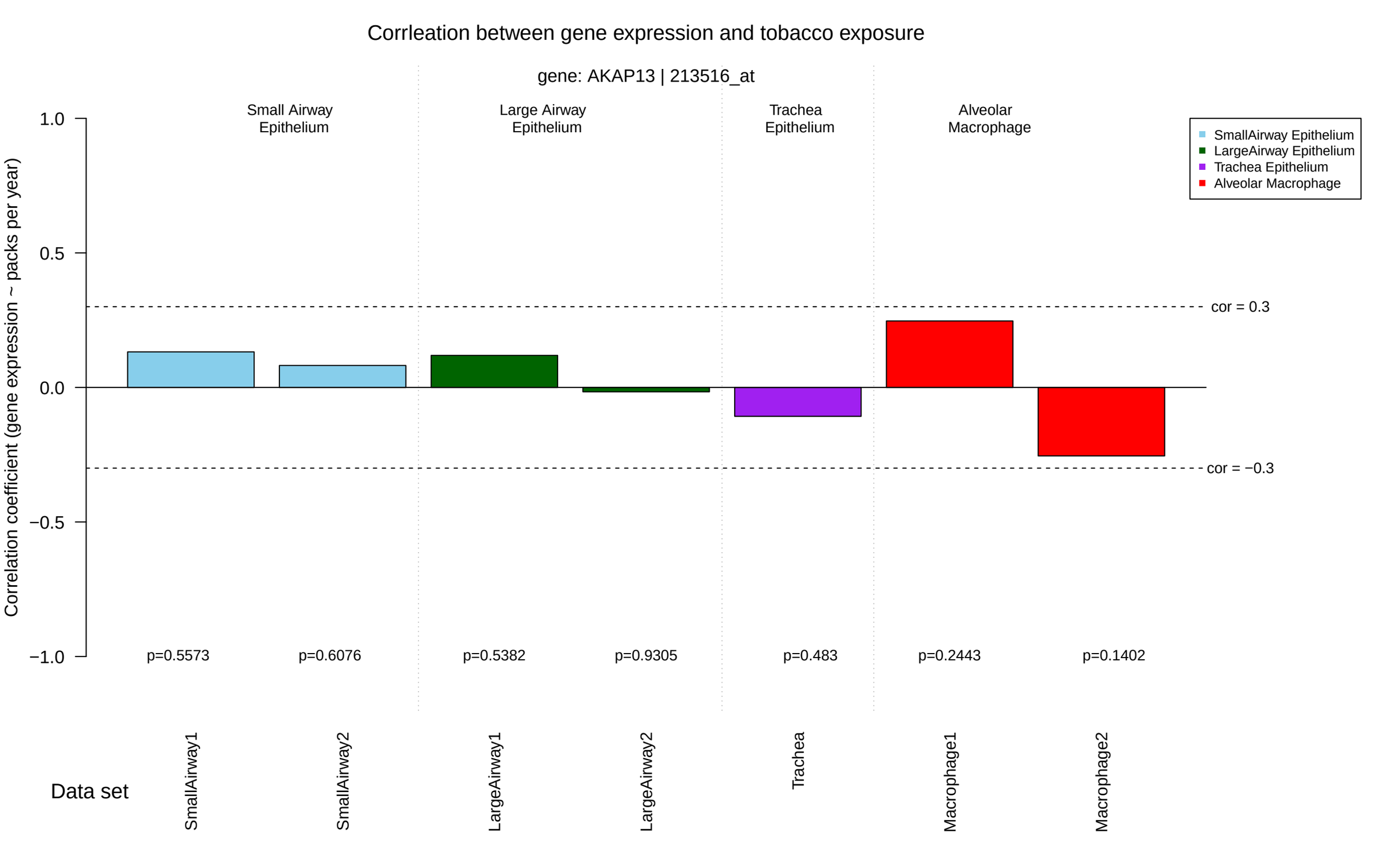

set: LargeAirway1 (p=0.2112)
AKAP13 | 213516_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6978)
AKAP13 | 213516_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1021)
AKAP13 | 213516_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6431)
AKAP13 | 213516_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9224)
AKAP13 | 213516_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP13 | 213516_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5573
p=0.6076
p=0.5382
p=0.9305
p=0.483
p=0.2443
p=0.1402
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2